ASEN 3802 Lab 1: Structural Mechanics (Truss Lab)

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Smead Aerospace CU Boulder ASEN 3802: Aerospace Laboratory

I. INTRODUCTION

In Part 1 of this study, the structural response of a 16-bay truss is modeled using an equivalent simply supported Euler–Bernoulli beam. The truss is idealized as a beam whose flexural rigidity is defined by a constant Young's modulus and an effective area moment of inertia.

The objective of this part is to use analytical methods to calculate the equivalent beam's structural response under several loading conditions. The area moment of inertia for the composite cross-section is determined using the parallel axis theorem to subsequently analyze three different loading conditions on the beam. The three conditions are a centered point load, an off-centered point load, and two symmetrically placed point loads. For each case, internal bending moments are obtained from section cuts and static equilibrium and are integrated to determine slope and deflection. Boundary and continuity conditions are applied to solve for the constants of integration. The resulting moment distributions are used to compute mid-span deflection and bending stress at the location of the load cell. These derivations establish the theoretical predictions that will later be compared with experimental measurements.

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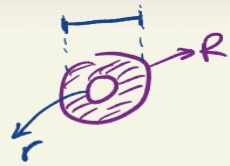
Part 1 Task 1

Part I (Task 1)

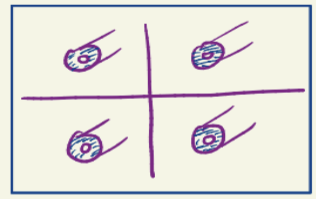
Fig. 2 Truss member (strut) cross section schematics. End fitting details (threads and joint adapters) not shown.



$$I = \frac{\pi}{4}(R^4 - r^4)$$

where R is outer radius

r is inner radius



$t = \frac{1}{16}$

$d = \frac{3}{8}$ so, $r = \frac{d}{2} = \frac{3}{16}$

$d = \frac{3}{8}$

$R = \frac{3}{8} \cdot \frac{1}{2} = \frac{3}{16}$

$R = \frac{3}{16}$ → "outer cylinder radius"

"Assume continuous rod"

$r = R - t$

$r = \frac{3}{16} - \frac{1}{16} = \frac{1}{8}$

$$I = \frac{\pi}{4}\left(\left(\frac{3}{16}\right)^4 - \left(\frac{1}{8}\right)^4\right) = 7.7897 \times 10^{-4} \text{ in}^4$$

$$A = \frac{\pi\left[\left(\frac{3}{8}\right)^2 - \left(\frac{3}{8} - \frac{2}{16}\right)^2\right]}{4}$$

$$= 0.061359 \text{ in}^2$$

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$$I = \bar{I} + Ad^2$$

$$= \bar{I} + (0.061359)\left(\frac{9.84252}{2}\right)^2$$

$$= 7.7897E-4 + (0.061359)\left(\frac{9.84252}{2}\right)^2$$

$$= 1.4868259 \text{ in}^4$$

$$\Sigma I = 4I = 4 \times 1.4868259 \text{ in}^4$$

$$= 5.947 (\text{in})^4$$

For task 1 the area moment of inertia was calculated by utilizing parallel axis theorem and summing up the MOI of each bar. Since the spacing between the bars and the axis was the same their MOI was the same so the MOI could be calculated by multiplying the MOI of one bar by 4. The final answer in inches to the fourth was 5.947.

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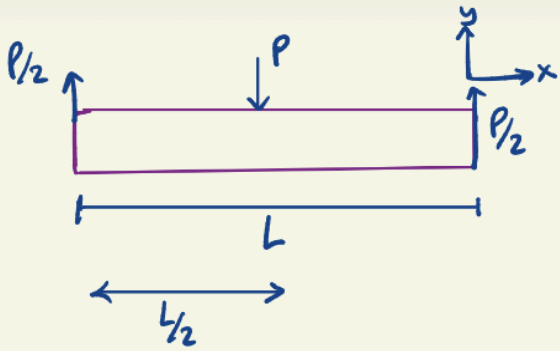
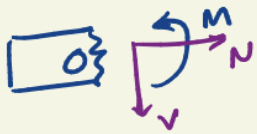
Part 1 Task 2

Task 2



$$\Sigma M_o = 0 = M - \frac{P}{2}x$$



$$\Sigma M_o = 0 = M - \frac{P}{2}x$$

$$M = \frac{P}{2}x$$

$$v = \frac{1}{EI}\iint M\,dx$$

$$\theta = \frac{1}{EI}\int M\,dx \Rightarrow \frac{1}{EI}\left(\frac{Px^2}{4} + C_1\right)$$

$$v = \frac{1}{EI}\left(\frac{Px^3}{12} + C_1x + C_2\right)$$

$x = \frac{L}{2}$ $\quad \theta = 0$ $\quad 0 = \frac{1}{EI}\left(\frac{PL^2}{16} + C_1\right)$

$x = 0$ $\quad v = 0$ $\quad C_1 = \frac{-PL^2}{16}$

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$$C_2 = 0$$

$$v = \frac{1}{EI}\left(\frac{Px^3}{12} + C_1 x\right)$$

$$v = \frac{1}{EI}\left(\frac{Px^3}{12} - \frac{P.L^2}{16}x\right)$$

$$x = \frac{L}{2}$$

$$v = \frac{1}{EI}\left(-\frac{PL^3}{48}\right)$$

$$v = \frac{1}{EI}\left(\frac{16Px^3 - 12PL^2x}{(16)(12)}\right)$$

$$x = L$$

$$v = \frac{1}{EI}\left(\frac{PL^3}{48}\right)$$

The derivation for the deflection of the beam with a single load at the center was done with one cut to the left side of the load p. After the cut, the moment about the cut was calculated and then integrated twice to derive the deflection expression. The boundary conditions used were one for which the angle was zero at half the length, and at x=0 the defection is zero.

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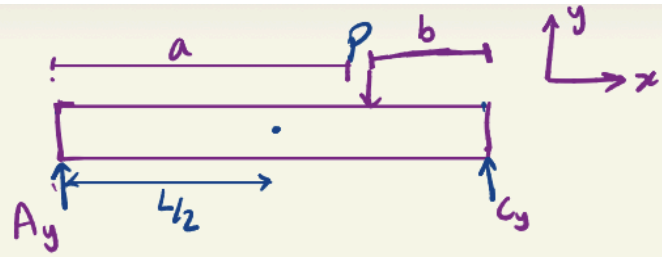
Part 1 Task 3

Task 3



$$x = a \quad , \quad L = a + b$$

$$b = L - a$$

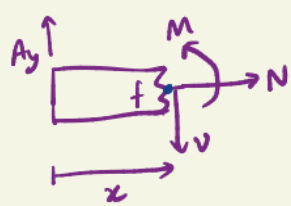
$$F_{net,y} = 0 = A_y + C_y - P$$

$$\Sigma M_o = 0 = -Pa + C_y L \Rightarrow C_y = \frac{Pa}{L}$$

$$0 = A_y + \frac{Pa}{L} - P \Rightarrow A_y = P - \frac{Pa}{L}$$

$$= P\left(1 - \frac{a}{L}\right)$$

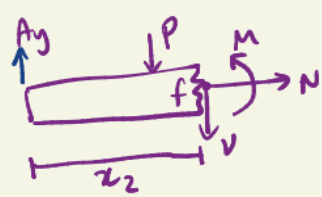
1



$$\Sigma M_f = 0 = M - A_y x$$

$$M = A_y x$$

2



$$\Sigma M_f = 0 = M - A_y x_2 + P(x_2 - a)$$

$$M = A_y x_2 - P(x_2 - a)$$

$$1: \theta_2 = \frac{1}{EI}\int M \, dx_2 = \frac{1}{EI}\left[\frac{A_y x_2^2}{2} - P\left(\frac{x_2^2}{2} - a x_2\right) + C_3\right]$$

$$v_2 = \frac{1}{EI}\left[\frac{A_y x_2^3}{6} - P\left(\frac{x_2^3}{6} - \frac{a x_2^2}{2}\right) + C_3 x_2 + C_4\right]$$

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I.C.

1: $x = 0$; $v_1 = 0$

$x = a$; $\theta_1 = \theta_2$

2: $x_2 = a$; $v_2 = v_1$

$x_2 = L$; $v_2 = 0$

1: $x = 0$; $v_1 = 0$ $\qquad C_2 = 0$

$x = a$; $v_1 = 0$ $\qquad \theta_2 = \frac{1}{EI}\left(\frac{A_y a^2}{2} + C_1\right)$

$$\theta_2 EI - \frac{A_y a^2}{2} = C_1$$

2: $$\theta_2 = \frac{1}{EI}\left[\frac{A_y a^2}{2} - P\left(\frac{a^2}{2} - a^2\right) + C_3\right]$$

$$= \frac{1}{EI}\left[\frac{A_y a^2}{2} + P\left(\frac{a^2}{2}\right) + C_3\right]$$

$$\theta_2 EI = C_3 + \frac{A_y a^2}{2} + \frac{Pa^2}{2}$$

$$C_1 = C_3 + P\frac{a^2}{2}$$

$x = a$; $v_2 = v_1$

$$v_1 = \frac{1}{EI}\left[\frac{A_y a^3}{6} - P\left(\frac{a^3}{3} - \frac{a^3}{2}\right) + C_3 a + C_4\right]$$

$x = L$; $v_2 = 0$

$$0 = \frac{1}{EI}\left[\frac{A_y L^3}{6} - P\left(\frac{L^3}{6} - \frac{aL^2}{2}\right) + C_3 L + C_4\right]$$

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$$v_1 = \frac{1}{EI}\left[\frac{A_y x^3}{6} + C_1 x\right]$$

$$= \frac{1}{EI}\left[\frac{A_y x^3}{6} + \left(C_3 + P\frac{a^2}{2}\right)x\right]$$

$$x = a \quad ; \; v_1 = v_2$$

$$v_1 = \frac{1}{EI}\left[\frac{A_y a^3}{6} + C_3 a + \frac{P a^3}{2}\right]$$

$$\cancel{\frac{A_y \cdot a^3}{6}} + \cancel{C_3 \cdot a} + \frac{P a^3}{2} = \cancel{\frac{A_y \cdot a^3}{6}} - P\left(\frac{a^3}{6} - \frac{a^3}{2}\right)$$

$$+ \cancel{C_3 \cdot a} + C_4 - C_3 a$$

$$\frac{P a^3}{2} = -P\left(\frac{a^3}{6} - \frac{a^3}{2}\right) + C_4$$

$$C_4 = \frac{P a^3}{2} + P\left(\frac{a^3}{6} - \frac{a^3}{2}\right) = \boxed{P\frac{a^3}{6}}$$

$$0 = \frac{A_y L^3}{6} - P\left(\frac{L^3}{6} - \frac{a L^2}{2}\right) + C_3 L + P\frac{a^3}{6}$$

$$C_3 = \frac{-\left[\frac{A_y \cdot L^3}{6} - P\left(\frac{L^3}{6} + \frac{a L^2}{2}\right) + P\frac{a^3}{6}\right]}{L}$$

$$= \frac{-A_y L^2}{6} + \frac{P L^2}{6} + \frac{P a L}{2} - P\frac{a^3}{6L}$$

$$A_y = P\left(1 - \frac{a}{L}\right) = P - \frac{Pa}{L}$$

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$$= -\frac{PL^2}{6} + \frac{PaL}{6} + \frac{PL^2}{6} + \frac{3PaL}{6} - \frac{Pa^3}{6L}$$

$$= \frac{2PaL}{3} - \frac{Pa^3}{6L}$$

$$\boxed{v_1 = \frac{1}{EI}\left[\frac{A_y x^3}{6} + \left(C_3 + P\frac{a^2}{2}\right)x\right]}$$

$$\boxed{v_2 = \frac{1}{EI}\left[\frac{A_y x_2^3}{6} - P\left(\frac{x_2^3}{6} - \frac{a x_2^3}{2}\right) + C_3 x_2 + P\frac{a^3}{6}\right]}$$

For task 3 two cuts were needed one before and after the load, and four boundary conditions: one at x_1=0 then v_1 = 0, at x_1=a then Θ_1 =0, at x_2= a then v_2 = v_1, and at x_2=L then v_2=0.

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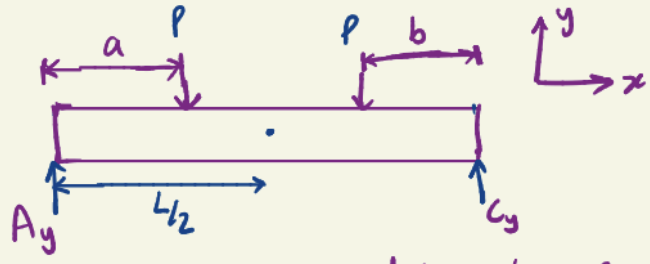
Part 1 Task 4

Task 4



Consider a beam is supported by two symmetric loads P on either side at distance a from left and distance b from right as shown in the figure.

$$A_y = C_y = P$$

For, $0 \leq x \leq a$ (from left)

Bending moment

$$M(x) = Px$$

For $a \leq x \leq (L-a)$ (middle section)

$$M(x) = Pa$$

$$V(x) = \frac{1}{EI} \iint M(x)\, dx$$

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$$\theta = \frac{1}{EI}\int M(x)\,dx = \frac{1}{EI}\left[\frac{Px^2}{2} + C_1\right]$$

$$\theta' = V(x) = \frac{1}{EI}\left[\frac{Px^3}{6} + C_1 x + C_2\right]$$

Using $V(0) = 0$ at pins

$$C_2 = 0$$

Slope at mid span $= 0$

@ $x = L/2$

$$\theta = 0$$

$$\frac{Px^2}{2} + C_1 = 0$$

$$\frac{P(L/2)^2}{2} + C_1 = 0$$

$$C_1 = \frac{-PL^2}{8}$$

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For, $0 \le x \le a$ (left section)

$$\boxed{v(x) = \frac{P}{6EI}\left(x^3 - \frac{3L^2}{4}x\right)}$$

For, $x = L/2$ (middle section)

$a \le x \le (L-a)$ if $a \le L/2$

$M(x) = Pa$

$$v(x) \frac{Pa}{24EI}(3L^2 - 4a^2)$$

$$\boxed{v(x) = \frac{Pa}{24EI}(3L^2 - 4a^2)}$$

For $(L-a) \le x \le L$ (Right section)

$M(x) = P(L-x)$

$$\theta = \frac{1}{EI}\int M(x)\,dx = P\left(Lx - \frac{x^2}{2}\right) + C_3$$

$$v(x) = \frac{1}{EI}\left[P\left(\frac{Lx^2}{2} - \frac{x^3}{6}\right) + C_3 x + C_4\right]$$

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@ $x = L$ (at pins)

$$v(0) = 0$$

$$0 = P\left(\frac{L^3}{2} - \frac{L^3}{6}\right) + C_3 L + C_4$$

$$0 = \frac{PL^3}{3} + C_3 L + C_4$$

$$\boxed{v(x) = \frac{P(L-x)}{6EI}\left(2aL - a^2 - (L-x)^2\right)}$$

Expected stress at inline load cell:

$$\sigma = \frac{M_{middle} \cdot c}{I}$$

$$\boxed{\sigma = \frac{Pac}{I}}$$

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For task 4 three sections were analyzed and the moments integrated to get the deflection and angle equations.

Variables	**SI Units**	**Standard US Engineering Units**
P (Load)	**N**	**lbf**
L, a, b, x	**m**	**ft or inch**
Young's Modulus (E)	**Pa or N/m^2**	**lbf/in^2 or (psi)**
Moment of Inertia (I)	**m^4 or mm^4**	**in^4**
Moment (M)	**Nm**	**ft-lb**

II. CONCLUSION

In conclusion the derivations for the different loadings of the beam followed the same process. First take section cuts at points of interest around the loads and then determine boundary conditions. Once the moment is calculated from the section cuts then it is integrated twice to get deflection and once to get theta. The last step is to solve for constants, for that plug in the boundary conditions and sometimes plug in the equations at different conditions into one another, then rearranging for the constants was, this was evident especially in task 3.

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Contribution Report - Structural Mechanics

Ritik Sarraf: I've contributed to all Part 1→ Task 1, Task 2, Task 3 and Task 4. I did Task 1 including diagrams, problem-solving, setting up equations and derivations and the same as follows for Task 4.

Quinn Jochim: Nothing on part 1, have been working on part 2 since the start.

Anton Pylypchenko: I wrote the introduction for part 1, but in terms of tasks I have been working on Part 3.

Harroop Sooch: I did the calculations for task 1-3, and did the formatting and writing in this document.

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